

Tomography

Ketil Hokstad, February 2016

Ketil Hokstad, January 2025

Things we have learnt so far

PSDM methods

- Kirchhoff migration
- Gaussian-beam migration
- Angle migration
- FK-migration (one-way equations)
- FD-migration (one-way equations)

Still to come

- Least-squares migration
- Reverse-time migration (RTM)

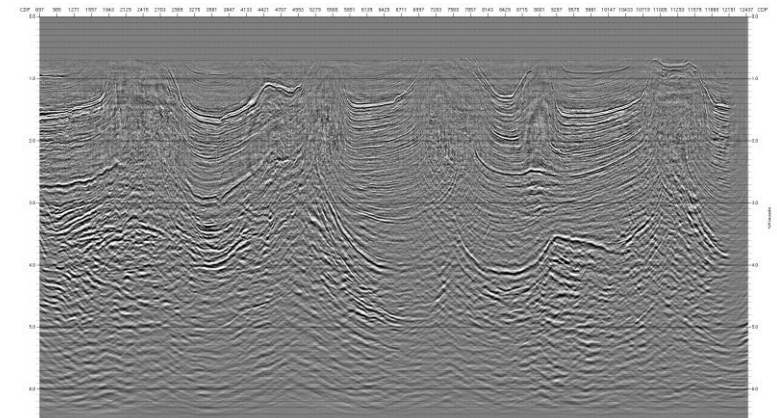
Next:

How to estimate velocity models

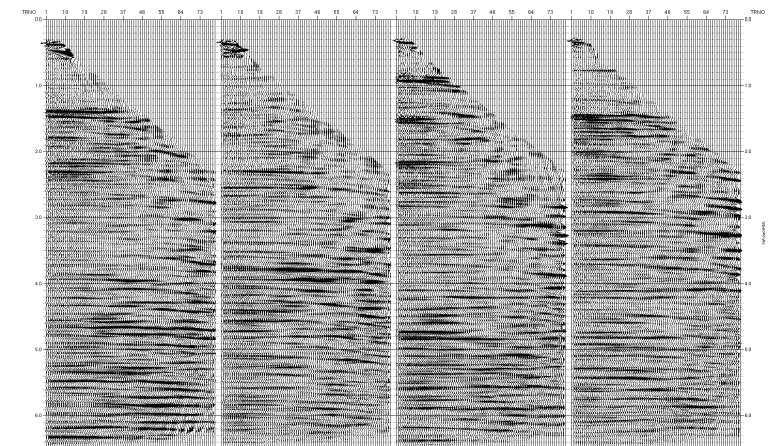
- Tomography

Later:

- Full-waveform inversion (FWI)
(later; non-linear inversion)



Common offset



Common image point

Reflection tomography

Tomo equation:

$$t = \int_{ray} s dl$$

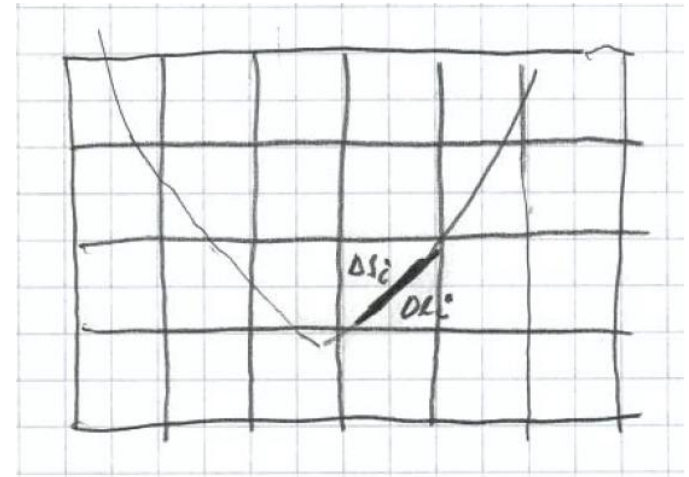
Isotropic: $s = |\mathbf{p}|$

Anisotropic: $s = 1/V$

Traveltime residual:

$$\Delta t = \int_{ray} \Delta s dl \simeq \sum_{ray} \Delta s_i \Delta l_i$$

Δt = change in traveltime due to a small change Δs in velocity model.



Linear system of equations:

$$\Delta \mathbf{t} = \mathbf{L} \Delta \mathbf{s}$$

Velocity analysis in (premigration) data domain

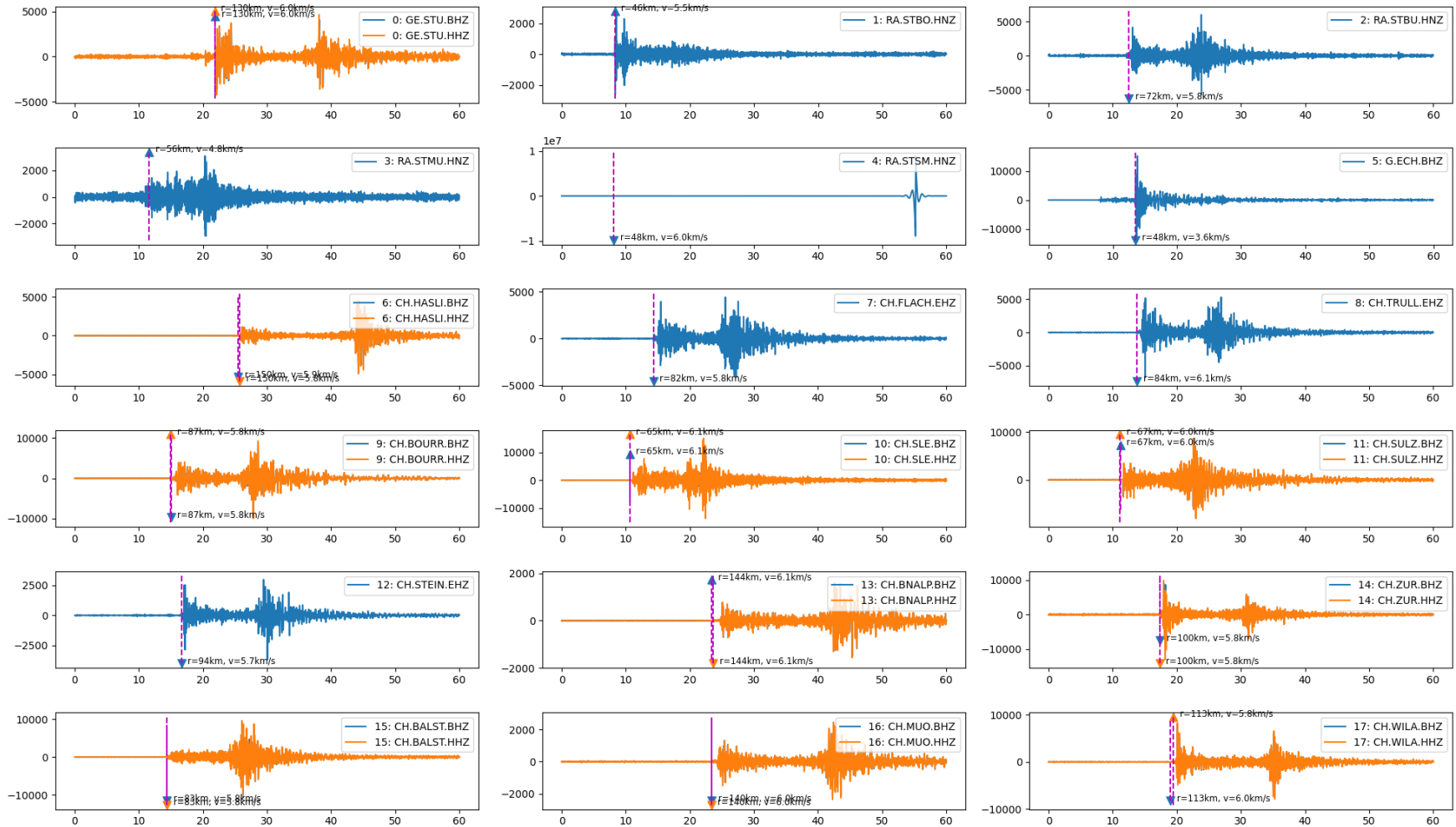
1. Sort to data common offset and pick traveltimes (reflection seismic)
2. Build starting model, incl reflectors
3. Ray trace => traveltimes
4. Compute $\Delta t = t_{pick} - t_{ray}$
5. Solve tomo equation $\Delta \mathbf{t} = \mathbf{L} \Delta \mathbf{s}$
6. Update velocity model

Disadvantage of this approach: Traveltime picking needed (using AI)

- Useful for earthquake tomography

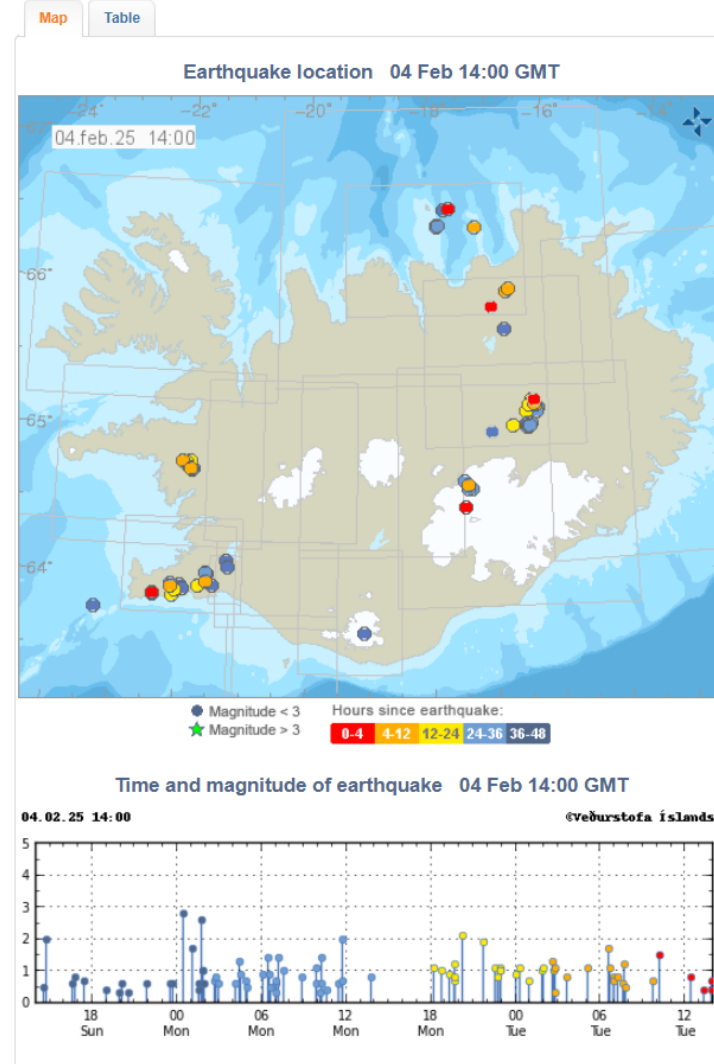
Earthquake seismograms and P-wave picks

0: Event 223, 2005-03-14T12:34:59.964999Z, ML=4.0



Iceland earthquake catalog (vedur.is)

Whole country - earthquakes during the last 48 hours (Preliminary results)



Whole country - earthquakes during the last 48 hours (Preliminary results)

Map Table

Earthquake count:

- Magnitude less than 1 in all: 53
- Magnitude 1 to 2 in all: 28
- Magnitude 2 to 3 in all: 5
- Magnitude more than 3 in all: 0
- Total: 86

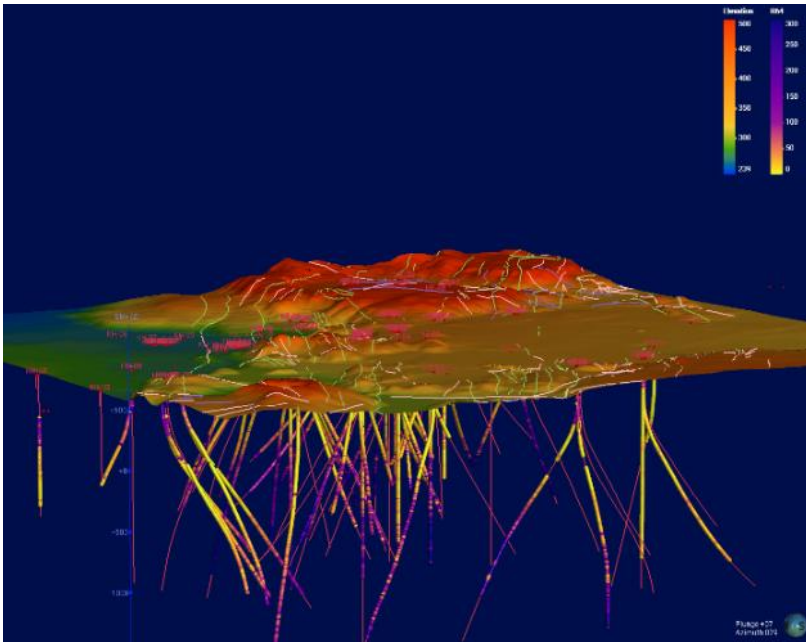
Earthquakes shown: All earthquakes Opna

Earthquake table

Date	Time	Latitude	Longitude	Depth	Magnitude	Quality	Location
Tuesday 04.02.2025	13:56:52	65.216	-16.352	5.2 km	0.7	53.54	4.7 km N of Herðubreið
Tuesday 04.02.2025	13:56:31	65.866	-16.985	4.5 km	0.4	68.33	1.7 km SW of Peistareykir
Tuesday 04.02.2025	13:23:37	63.889	-22.377	0.4 km	0.4	99.0	1.0 km S of Stóra-Skógfell
Tuesday 04.02.2025	12:23:07	64.502	-17.492	6.3 km	0.8	99.0	15.1 km NW of Grímsfjall
Tuesday 04.02.2025	10:12:48	66.544	-17.672	10.3 km	1.5	99.0	14.9 km E of Grímsey
Tuesday 04.02.2025	09:47:12	63.977	-21.558	7.3 km	0.7	99.0	4.4 km E of Bláfallaskáli
Tuesday 04.02.2025	07:46:54	65.982	-16.699	7.8 km	0.5	99.0	8.9 km WSW of Ásbyrgi
Tuesday 04.02.2025	07:43:02	65.191	-16.337	6.0 km	1.2	99.0	1.9 km NNE of Herðubreið
Tuesday 04.02.2025	07:37:07	64.646	-17.450	2.2 km	0.6	99.0	3.7 km E of Bárðarbunga
Tuesday 04.02.2025	07:13:42	63.890	-22.384	3.7 km	0.8	99.0	1.0 km SSW of Stóra-Skógfell
Tuesday 04.02.2025	07:00:21	65.964	-16.739	7.7 km	0.7	99.0	11.3 km WSW of Ásbyrgi
Tuesday 04.02.2025	06:57:28	65.187	-16.333	4.6 km	0.8	99.0	1.6 km NNE of Herðubreið
Tuesday 04.02.2025	06:43:05	63.935	-22.100	4.4 km	1.1	99.0	3.6 km E of Keilir
Tuesday 04.02.2025	06:36:28	66.415	-17.242	7.8 km	1.7	99.0	36.8 km ESE of Grímsey
Tuesday 04.02.2025	05:05:34	63.936	-22.100	4.3 km	1.1	99.0	3.6 km E of Keilir
Tuesday 04.02.2025	03:40:46	64.745	-21.866	1.1 km	0.8	99.0	21.9 km N of Borgarnes
Tuesday 04.02.2025	02:50:50	63.940	-22.112	1.2 km	0.3	99.0	2.9 km E of Keilir
Tuesday 04.02.2025	02:47:38	64.797	-22.007	9.3 km	1.1	99.0	27.9 km N of Borgarnes
Tuesday 04.02.2025	02:43:45	64.750	-21.894	7.7 km	1.0	99.0	22.4 km N of Borgarnes
Tuesday 04.02.2025	02:36:43	65.980	-16.696	7.0 km	1.3	99.0	8.8 km WSW of Ásbyrgi
Tuesday 04.02.2025	01:56:55	63.914	-22.043	4.1 km	1.1	99.0	3.2 km NNE of Krýsuvík

Hengill geothermal area

Hellisheiði and Nesjavellir power plants



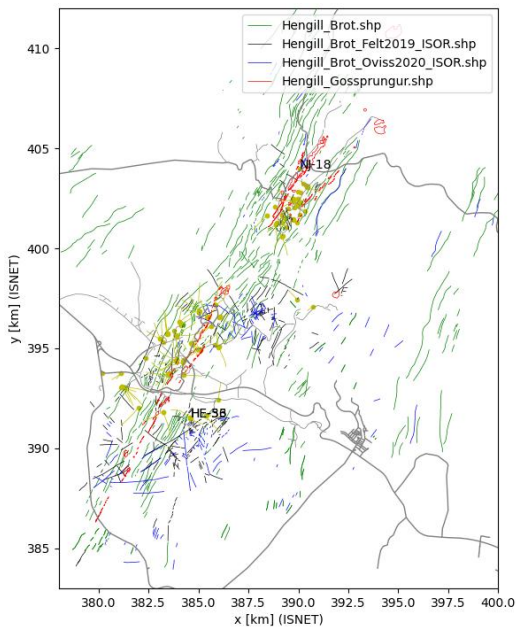
Hengill area. Leapfrog project
(from Reykjavik Energy (OR) and ISOR)



Hellisheiði powerplant (300 MWe)

Seismicity analysis and fault mapping

Goertz-Allmann et al. (2023)



Faults and fissures
(data from OR/ISOR)

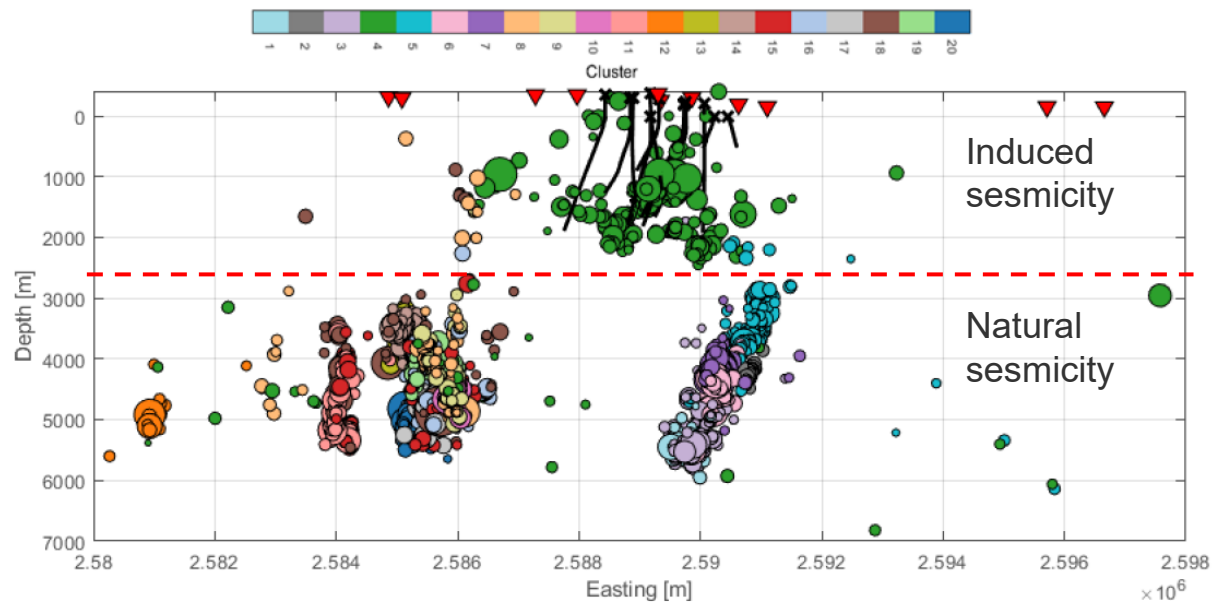


Figure 2: Depth cross-section of the microseismicity in the study area. Seismic events (circles) are colored by clusters of similar waveforms and scaled by magnitude. Production and injection wells from the Nesjavellir geothermal power plant are shown as black lines. Seismic stations are shown by inverted red triangles.

Earthquake tomography

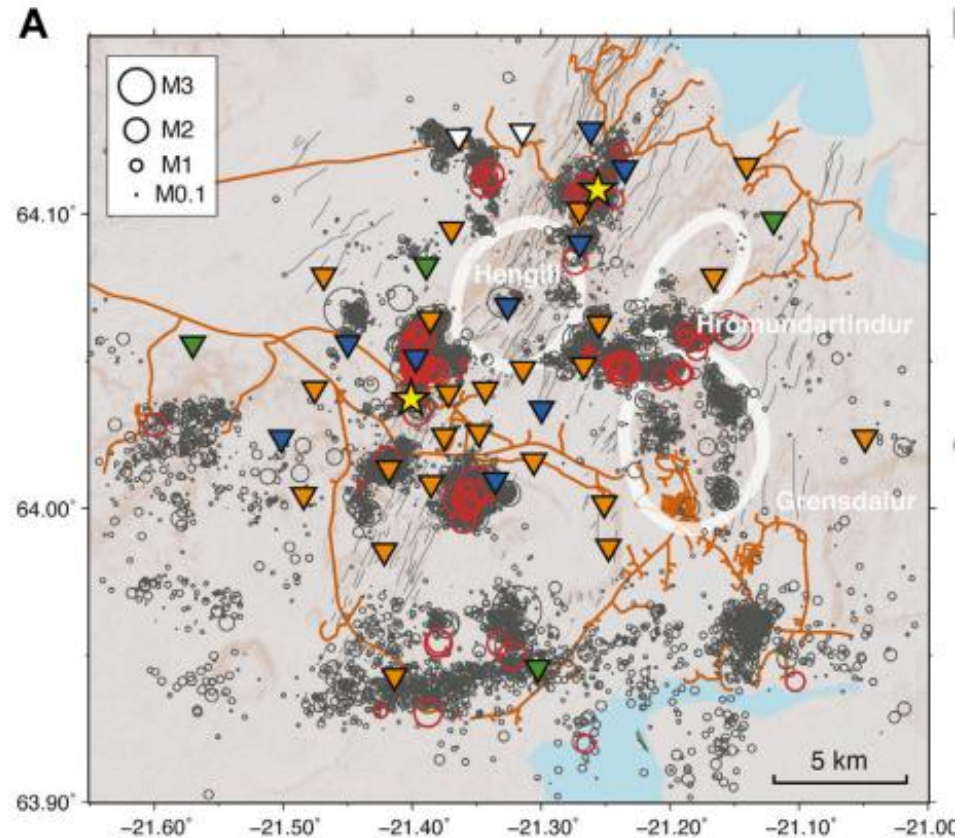
Hengill Volcanic Zone

Tomo equations:

$$T_P = \int_{ray} \frac{dl}{v_P}$$

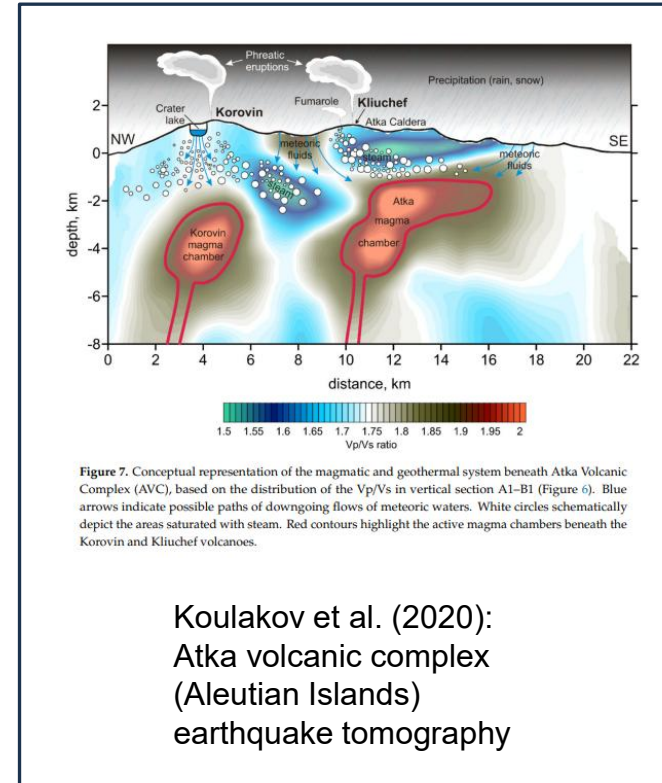
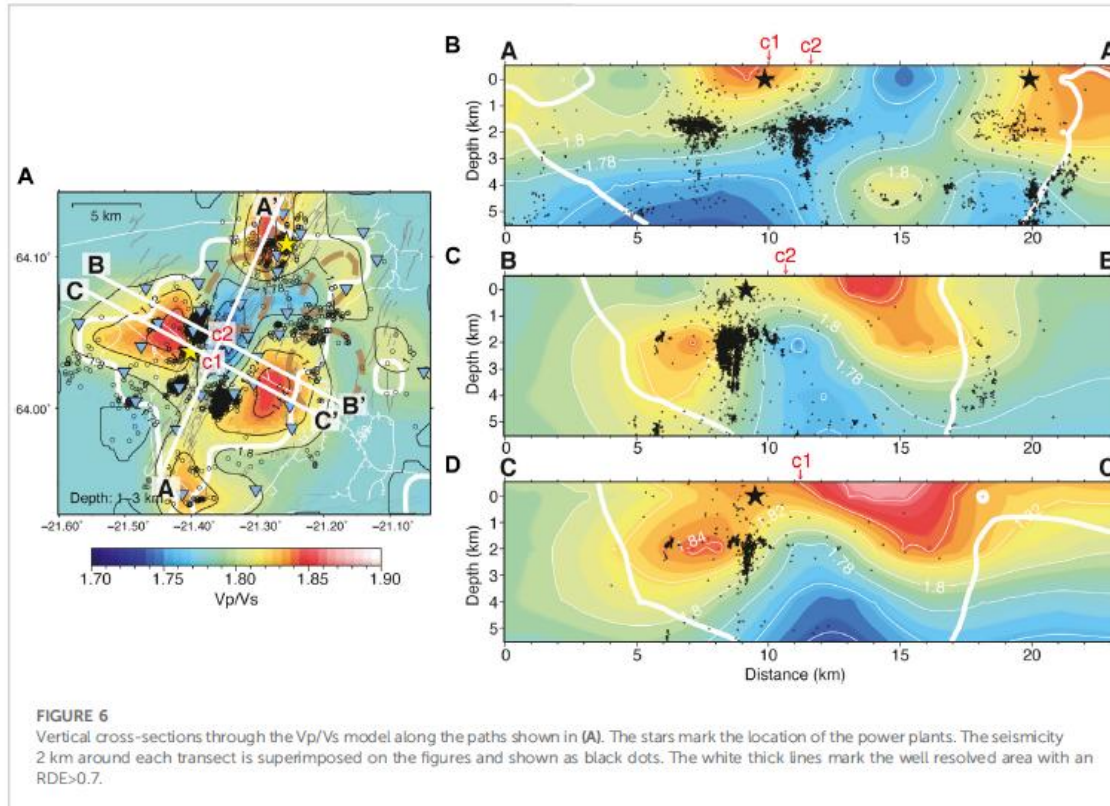
$$T_S - T_P = \int_{ray} \frac{dl}{v_P} \left(\frac{v_P}{v_S} - 1 \right)$$

- Regional study
- 15 permanent + 23 temporary seismometers
- ~6000 manually picked events from the full EQ catalogue
- 3D body wave tomography
- $dx=dy=3\text{km}$



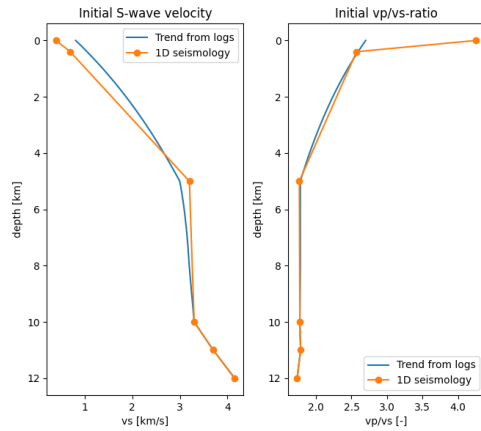
Earthquake tomography, v_P/v_S -ratio

Obermann et al. (2022)

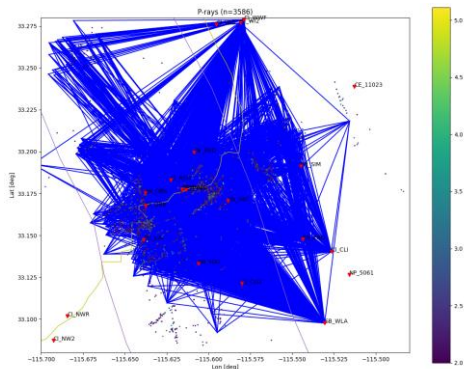
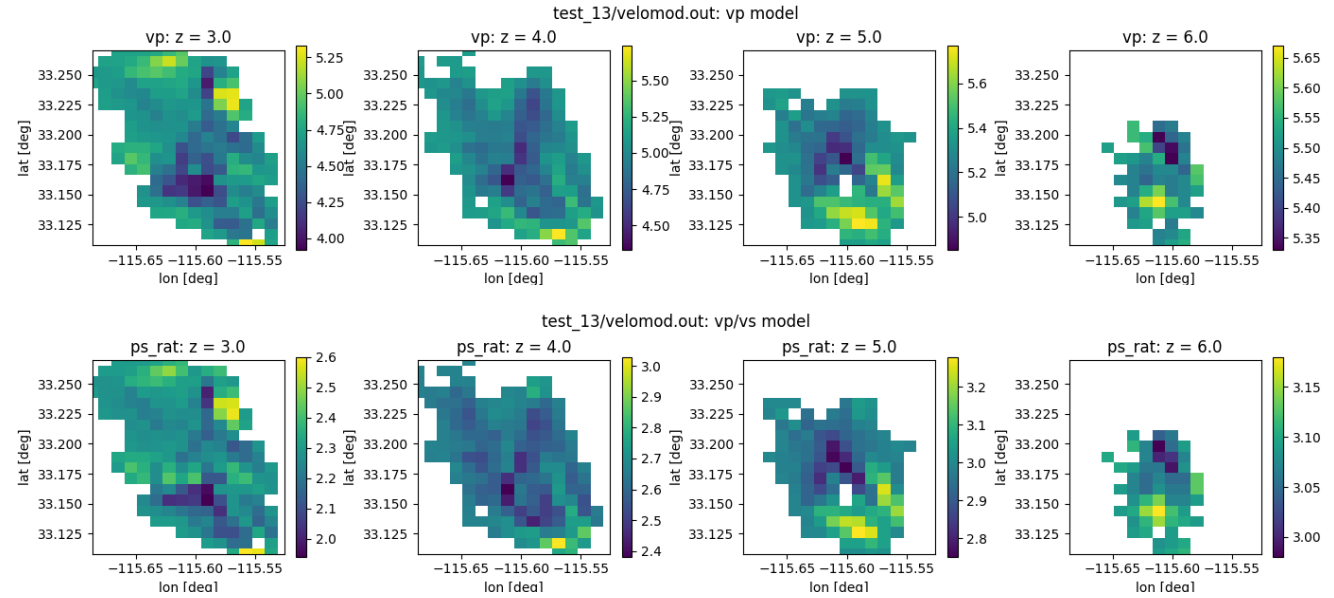


simulPS14 tomo code from USGS (Thurber, 1983; Evans et al., 1994)

simulPS14 tomography - SCEC data example



Initial model

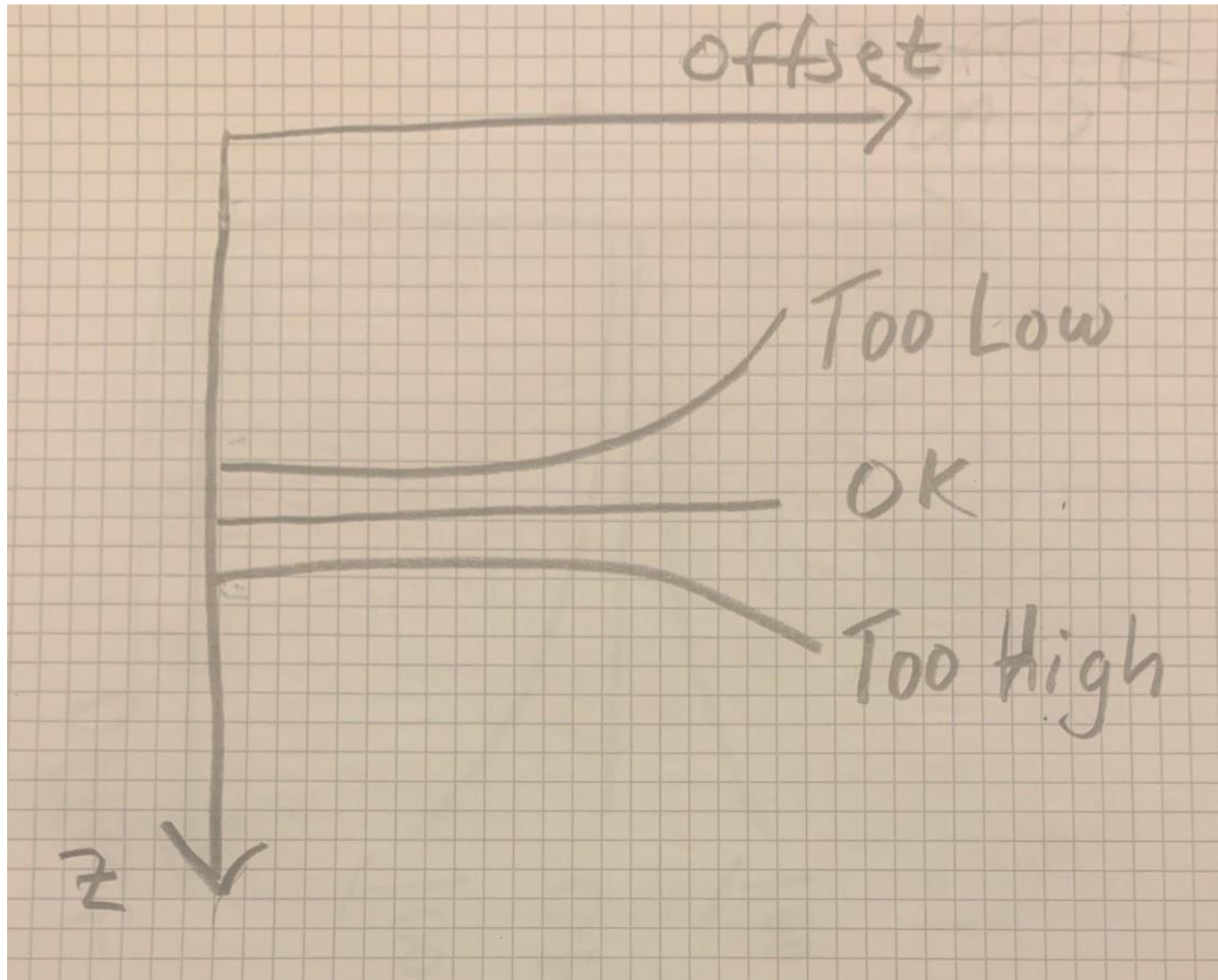


index	eid	etype	time0	lon0	lat0	depth0	mag	mtype	network	station	chan	lon	lat	elev	dist	phase	polarity	time	terr
0	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SSW	HHZ	-115.616	33.1775	-64	5.25	P	positive	1.179	0
1	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	CLI	EHZ	-115.527	33.1403	-92	7.58	P	negative	1.742	0
2	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	WI2	EHZ	-115.581	33.2768	-63	8.64	P	negative	1.882	0
3	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	IRS	EHZ	-115.429	33.1909	-43	12.77	P	positive	2.832	0
4	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	NW2	EHZ	-115.693	33.0873	-100	17.19	P	positive	3.782	0.01
5	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	FRK	EHZ	-115.638	33.4021	60	23.44	P	negative	4.162	0
6	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	ELR	EHZ	-115.833	33.1475	-97	25.53	P	positive	5.372	0
7	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SWS	HLZ	-115.8	32.9451	140	35.71	P	positive	6.382	0.01
8	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SU2	EHZ	-115.8	32.9448	160.8	35.74	P	positive	6.432	0
10	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	COK	EHZ	-115.728	32.8495	-50	41.76	P	positive	8.032	0
11	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	CRR	EHZ	-115.969	32.8868	66	51.15	P	positive	8.972	0
13	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	BC3	EHZ	-115.453	33.6548	1080	51.62	P	negative	8.982	0
15	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	COA	EHZ	-115.123	32.8634	36	55.61	P	negative	10.132	0.01
16	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	LTC	EHZ	-115.077	33.4938	275	55.97	P	negative	9.482	0
17	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	HAY	EHZ	-115.639	33.7074	413	56.78	P	negative	9.912	0
18	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	CTW	EHZ	-115.872	33.6773	504	60.19	P	negative	10.372	0
19	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	RUN	EHZ	-114.978	32.9722	116	60.23	P	positive	10.392	0
20	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SGI	EHZ	-115.726	32.6493	75	62.98	P	positive	10.562	0.01

Velocity analysis in postmigration (image) domain

1. Sort to data common offset
2. Perform common-offset migration
3. Sort to CIP gathers $I(\mathbf{x}, \mathbf{h})$
4. Compute $\Delta z_{RMO}(\mathbf{x}, \mathbf{h})$
5. Demigrate $\Delta z_{RMO}(\mathbf{x}, \mathbf{h})$ to get Δt
6. Solve tomo equation $\Delta \mathbf{t} = \mathbf{L} \Delta \mathbf{s}$
7. Update velocity model

Advantage: Picking RMO (deviation from flat gathers)
is easier than picking traveltimes in data



Multi-parameter tomography

Complete differential

$$\Delta T = \sum_j \frac{\partial T}{\partial m_j} \Delta m_j$$


Traveltime residual
(data domain)

Misfit may be measured in terms of Δz :

$$\Delta T = \frac{\partial T}{\partial z} \Delta z = p_z \Delta z$$
$$\Delta z = \frac{1}{p_z} \sum_j \frac{\partial T}{\partial m_j} \Delta m_j$$


Moveout residual
(image domain)

Anisotropy

Possible choices of model parameters:

$$\mathbf{m}(\mathbf{x}_k) = \{v_0, \delta, \epsilon\}$$

$$\mathbf{m}(\mathbf{x}_k) = \{v_2, \eta\}$$

$$v_2 = v_0 \sqrt{1 + 2\delta}, \text{ Interval NMO velocity}$$

$$\eta = \frac{\epsilon - \delta}{1 + 2\delta}, \text{ Alkalifah parameter}$$

$$\mathbf{m} = \begin{cases} v_0(\mathbf{x}_1) \\ \delta(\mathbf{x}_1) \\ \epsilon(\mathbf{x}_1) \\ v_0(\mathbf{x}_2) \\ \delta(\mathbf{x}_2) \\ \epsilon(\mathbf{x}_2) \\ \vdots \\ v_0(\mathbf{x}_N) \\ \delta(\mathbf{x}_N) \\ \epsilon(\mathbf{x}_N) \end{cases}$$

Linear system of equations:

$$\Delta \mathbf{d} = \mathbf{L} \Delta \mathbf{m}$$

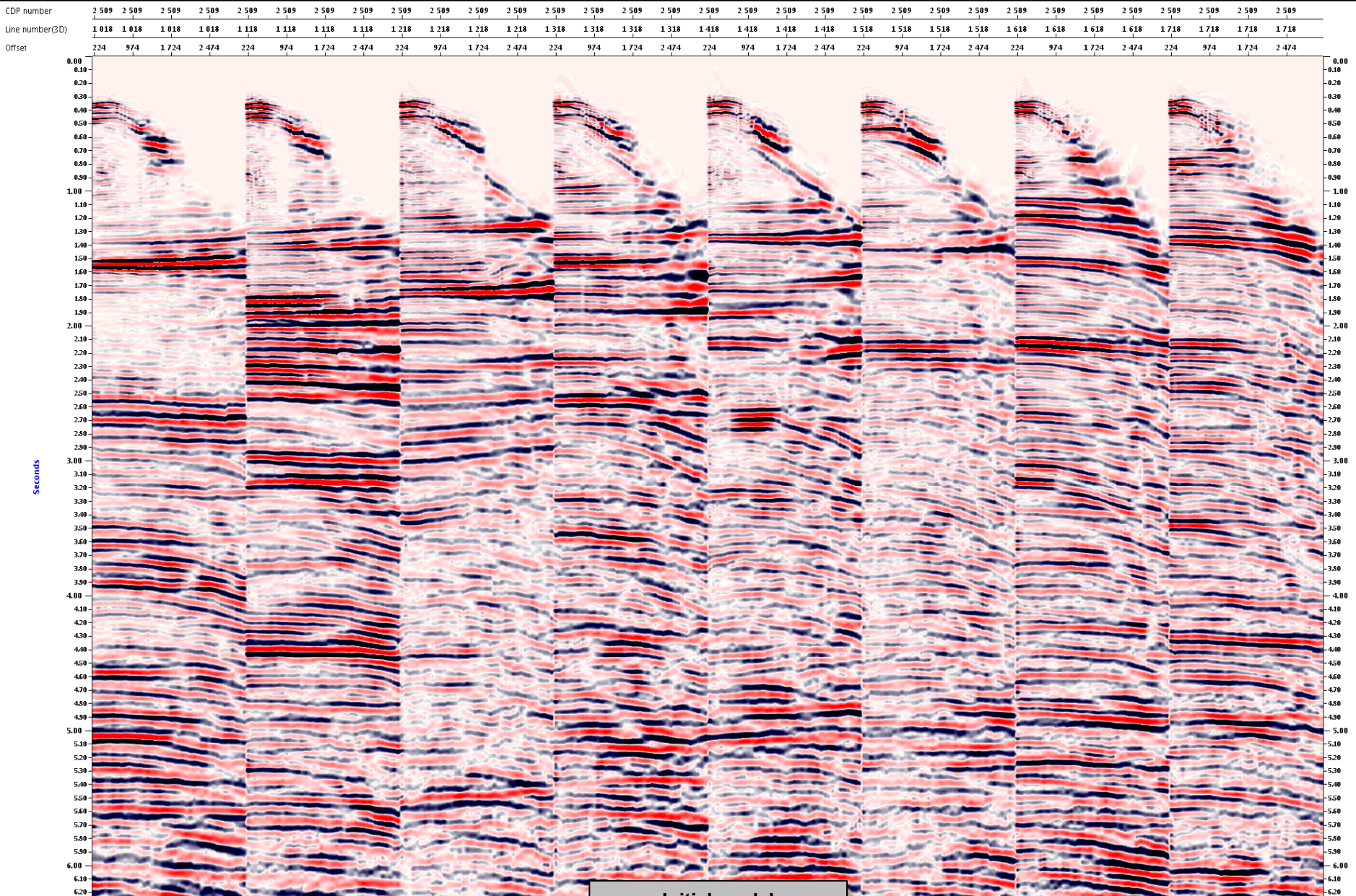
$$\Delta d_i = \Delta z_i$$

$$L_{ij} = \frac{1}{p_z} \frac{\partial T_i}{\partial m_j}$$

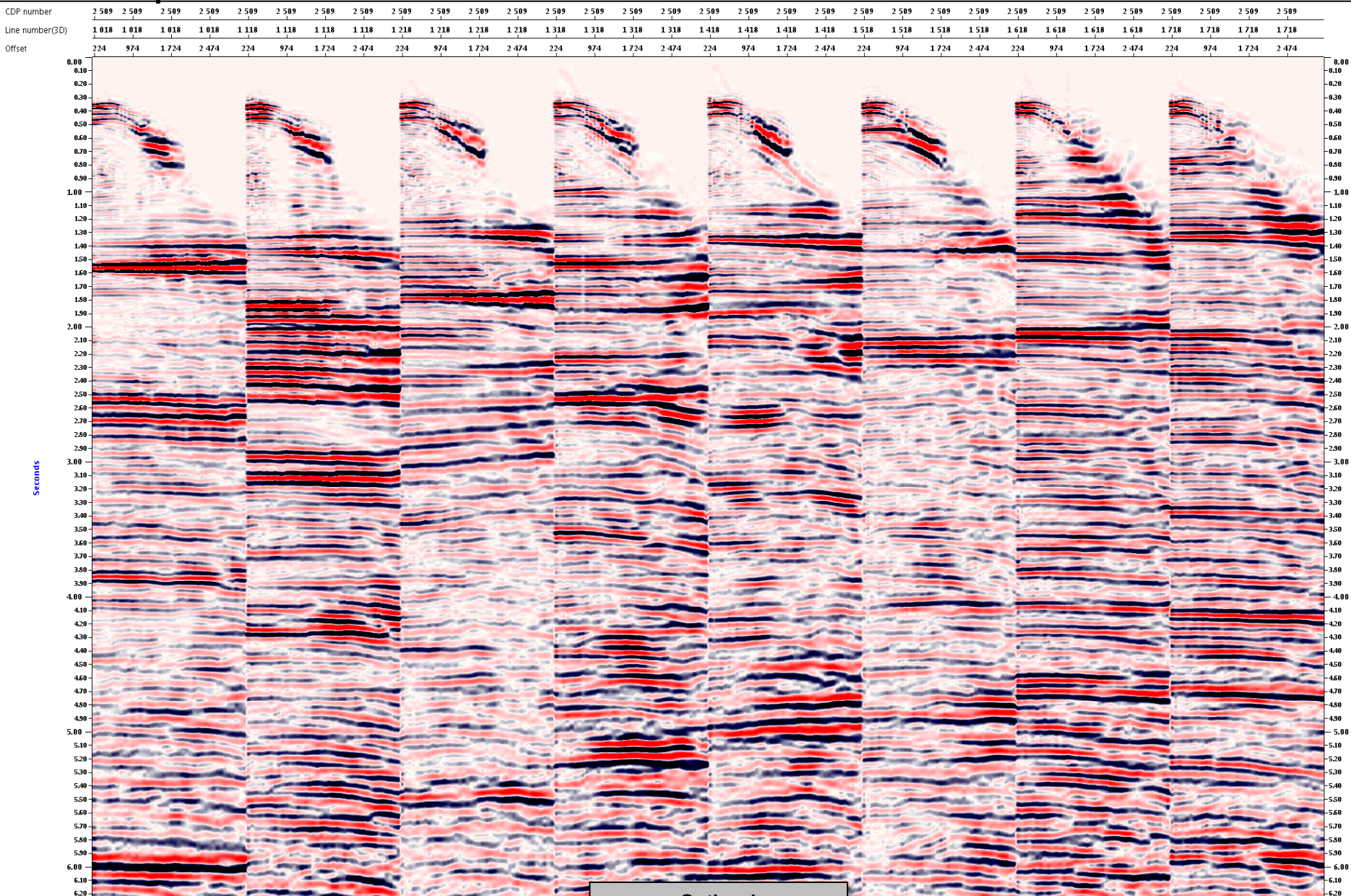
Python:

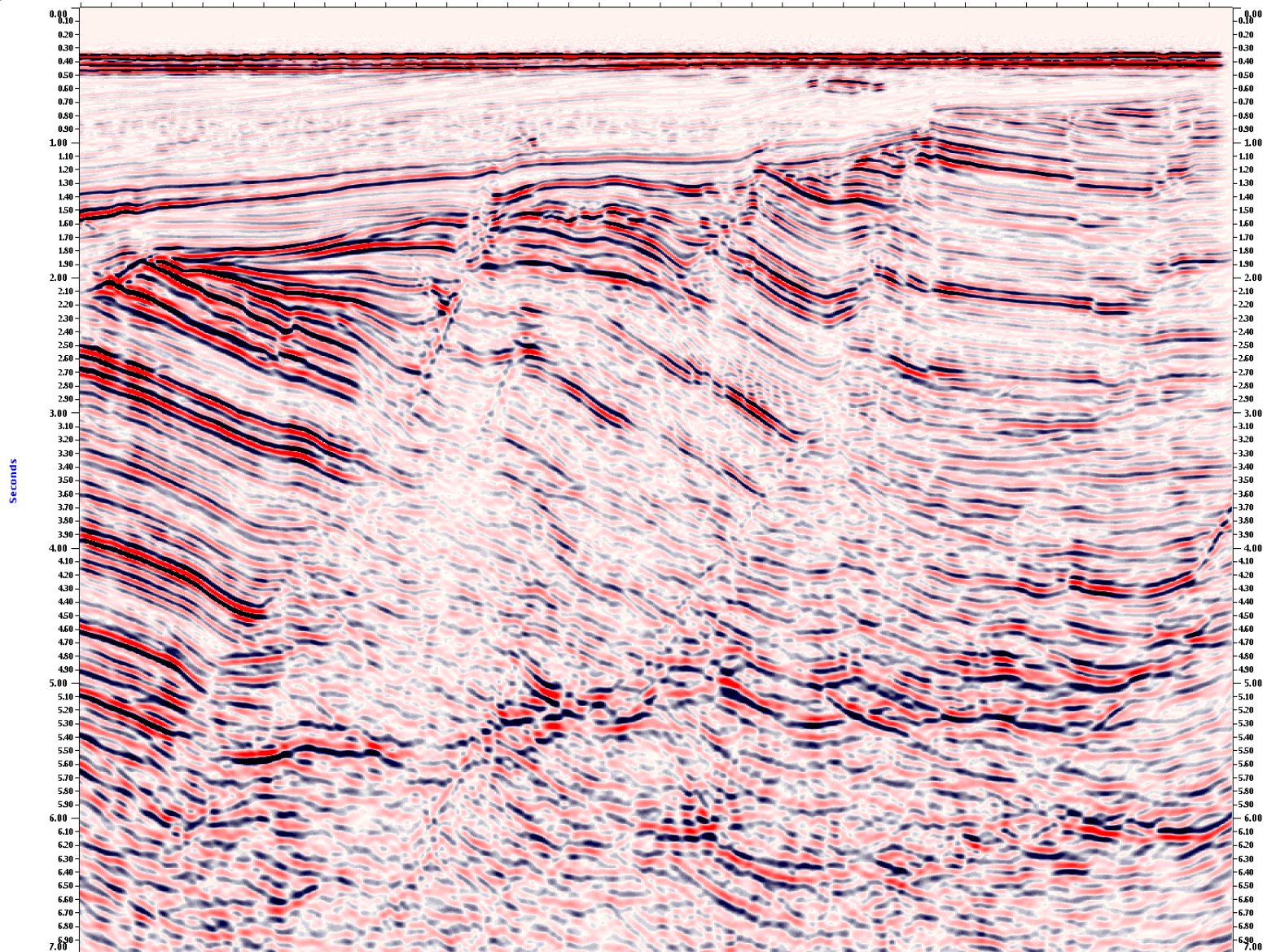
```
m.flatten()
m.ravel()
```

Initial model: CIP xl 2509

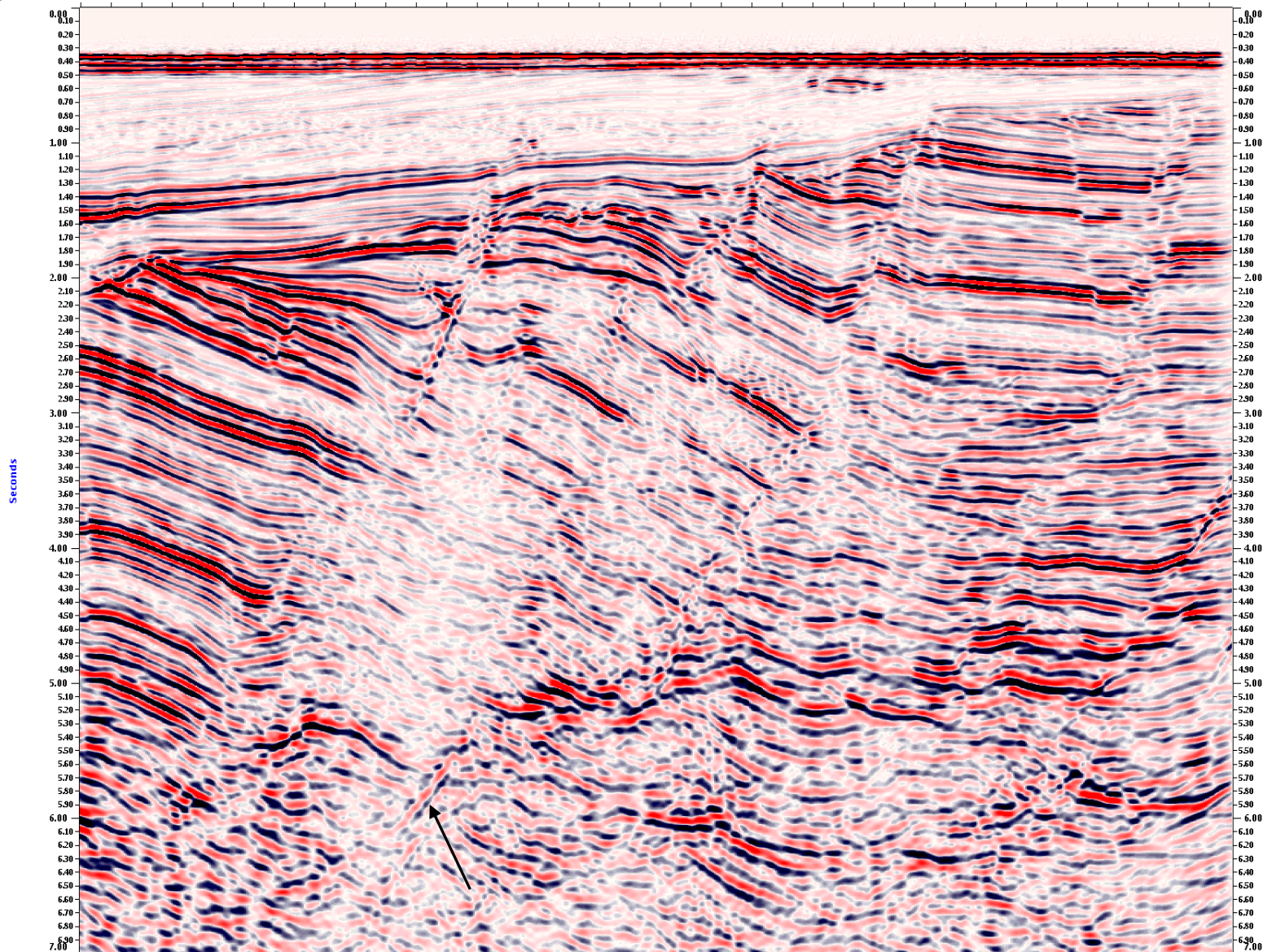


Updated model: CIP xl 2509



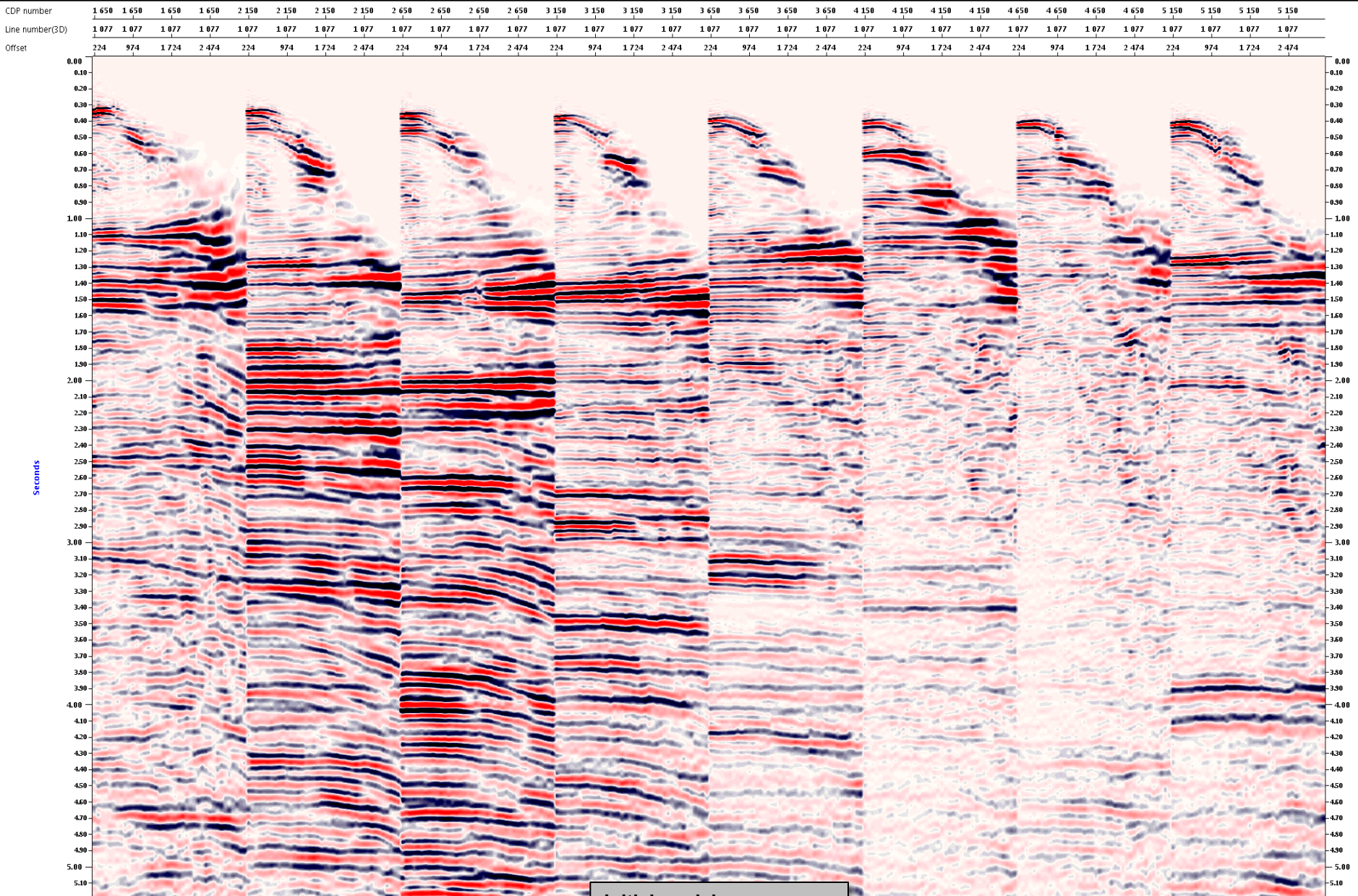
[illegible]

Initial model

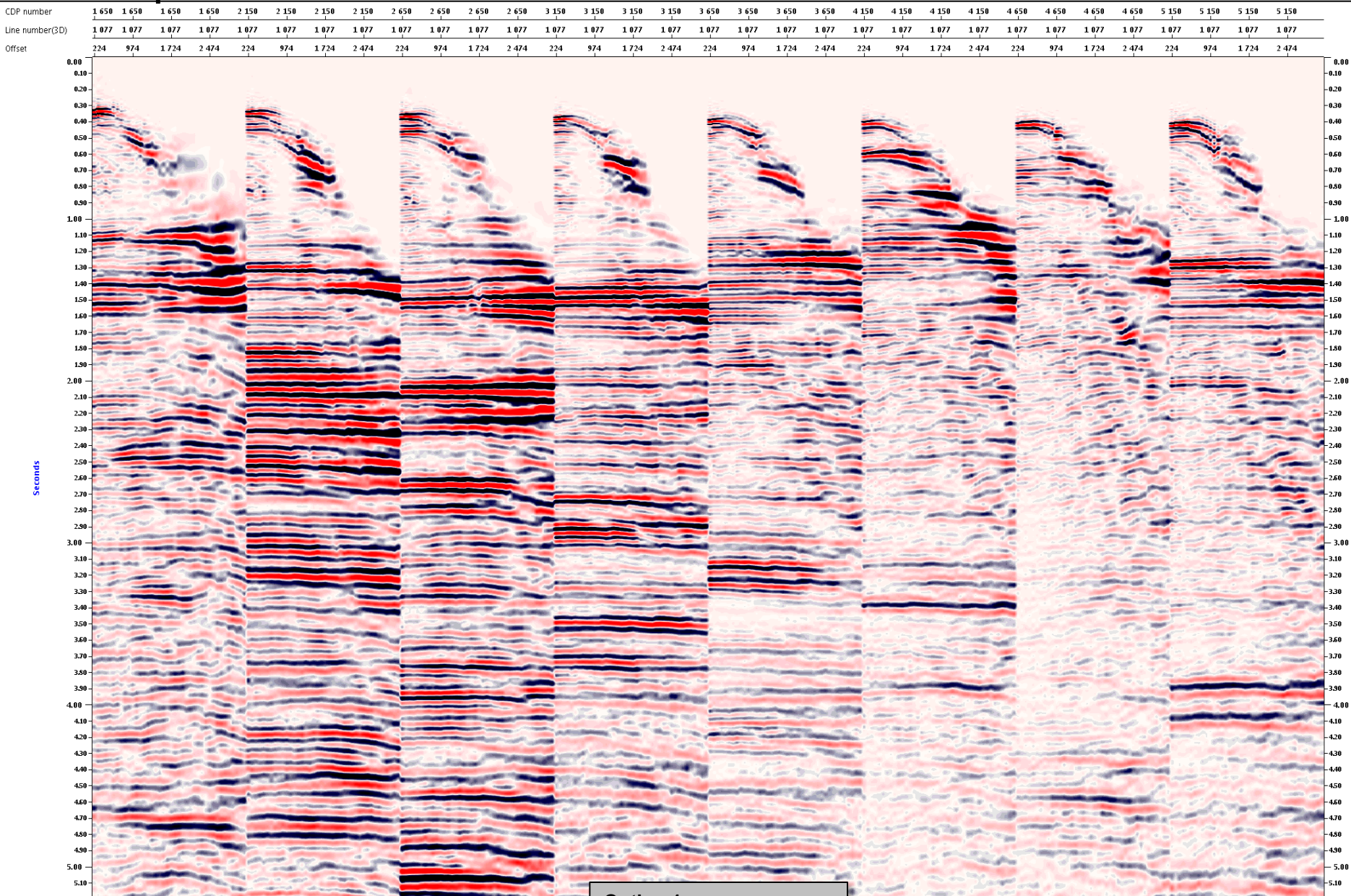
[illegible]

Option 1

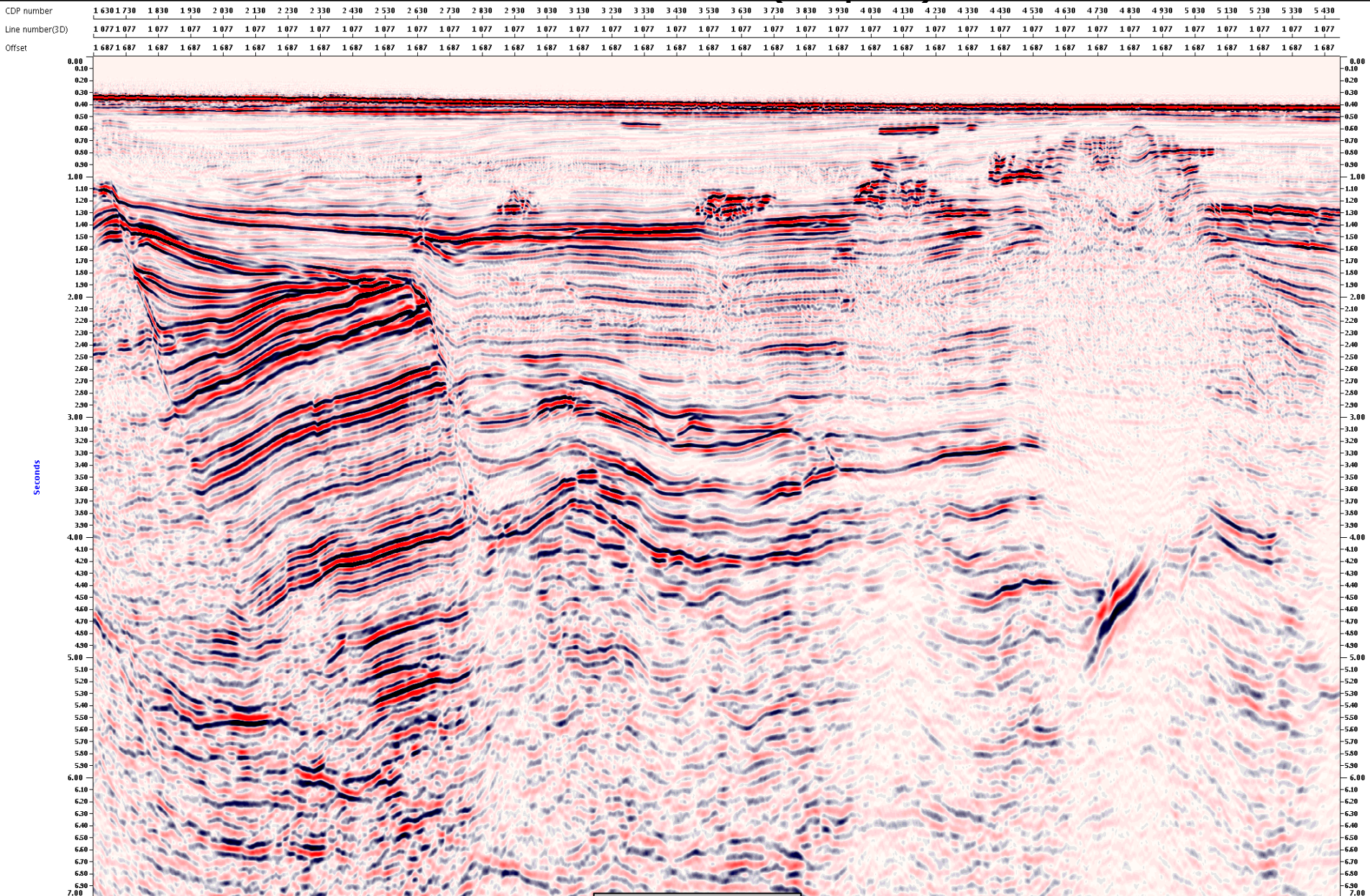
Initial model: CIP il 1077

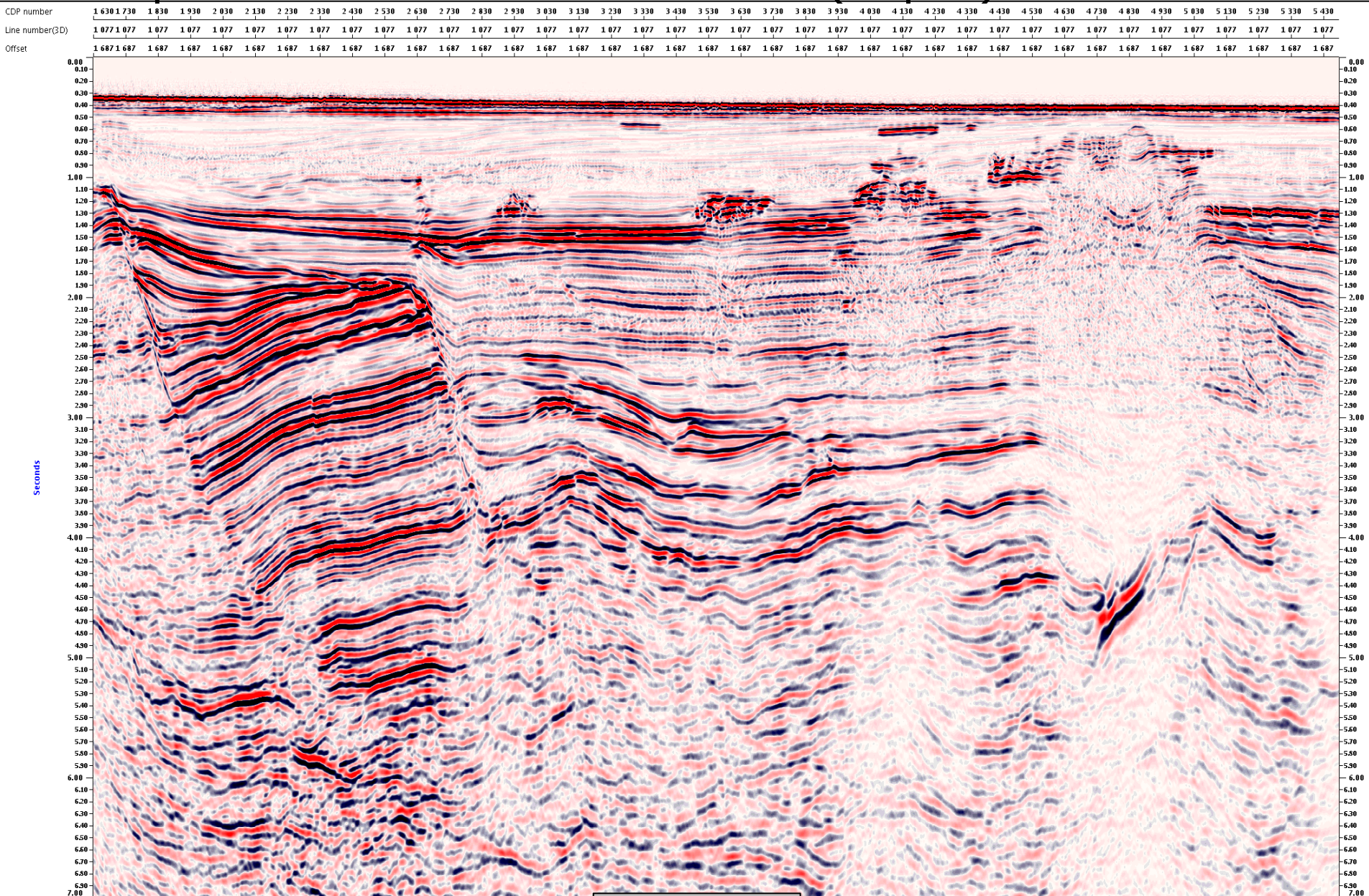


Updated model: CIP il 1077

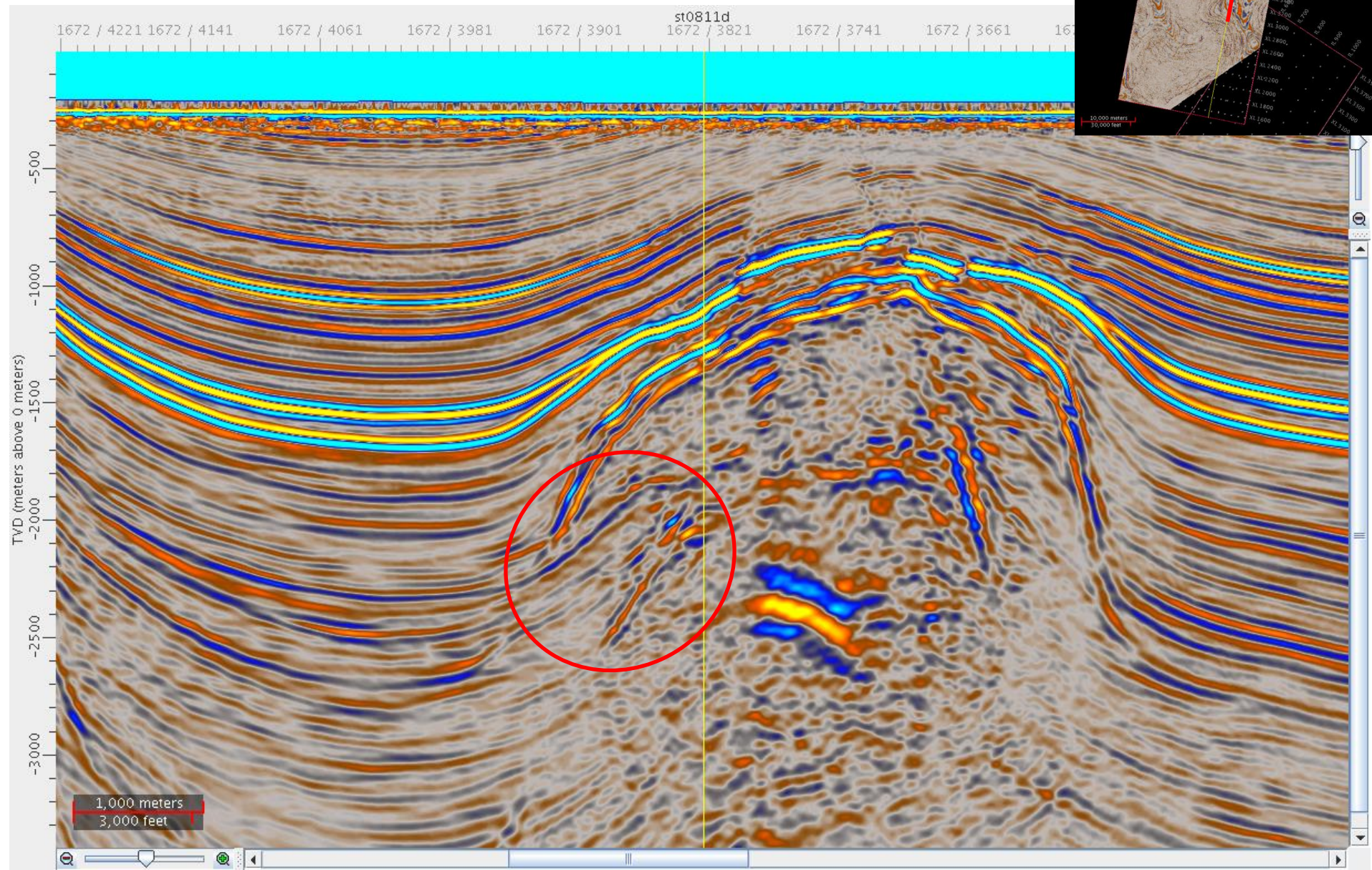


Initial model: Stack il 1077 (depth)

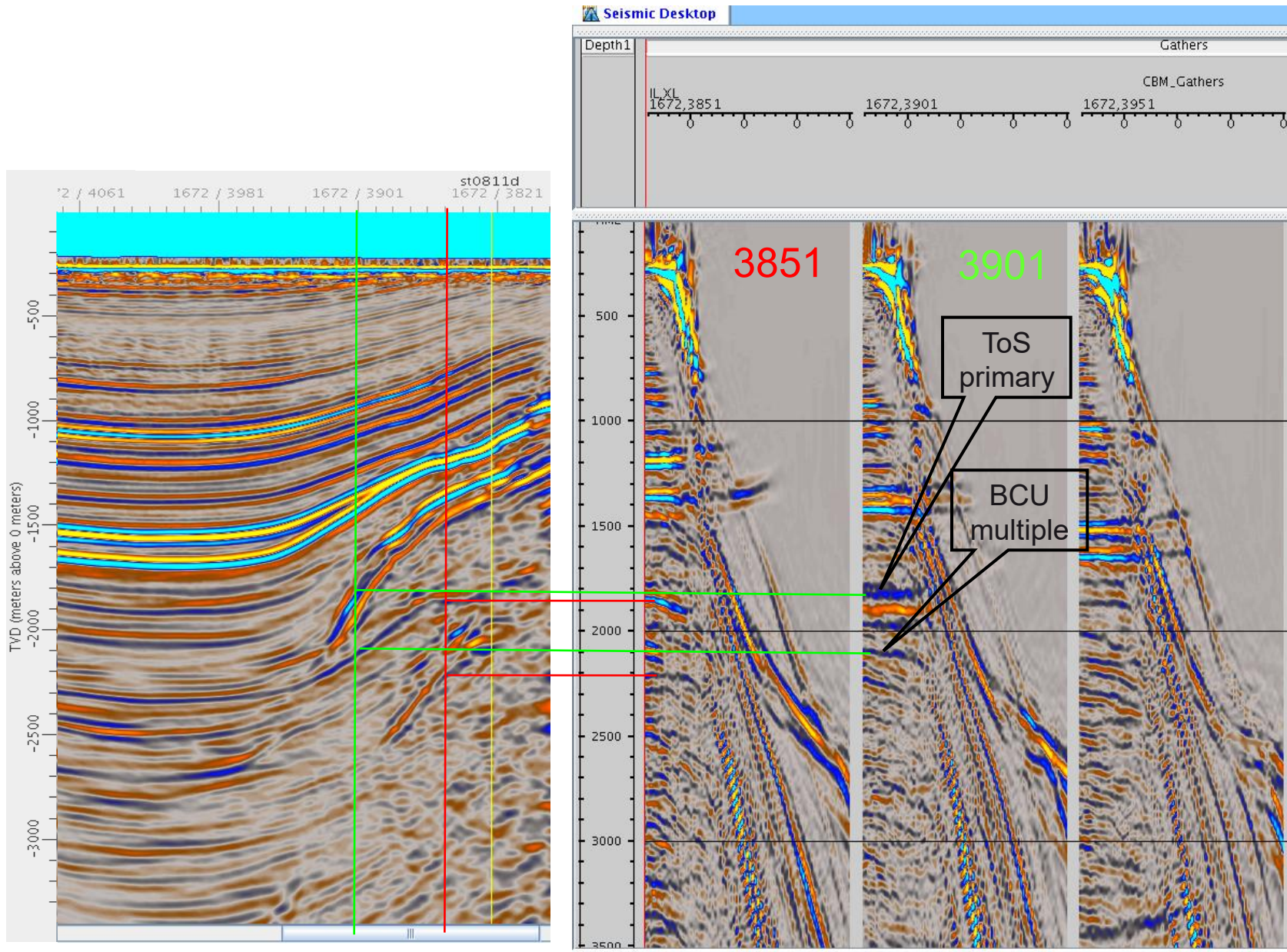




Final CBM IL 1672



IL 1672 Final CBM gathers before Radon



IL 1672 Final CBM gathers after Radon

